# Preface

Shanghai Zhenhua Heavy Industry Machinery Co.（ZPMC）is a famous heavy-duty machinery manufacturer specializing in container handling cranes. Its main products, Quayside Container Crane (QC) and Rubber Tyred Gantry Crane (RTG) sell across 63 countries and areas (including Chinese Taiwan and Hong Kong), covering major terminals all over the world. So far, ZPMC has captured more than 70% of the world market share in the industry. ZPMC has been endeavoring and growing in the severe domestic and overseas marketing competition and held in admiration for its high-technology, high-quality and high-reliability products, competitive prices, timely delivery and complete after-sale service to its customers. ZPMC has become a world-famous brand name in the industry.

This manual has been planned for many years. As an enterprise who manufactured more than 1250 Quayside Container Cranes and 2000 RTGs, it is ZPMC's obligation to produce a picture-illustrated, operable and practical maintenance manual for port customers. So far, ZPMC has just finished compiling this <Quayside Container Crane Configuration and Maintenance Manual> as some conditions are limited.

This manual focuses on configuration and maintenance tasks of quayside container crane especially on cleanliness, inspection fastening, adjustment, lubrication and replacement which are six basic aspects of maintenance job. The manual covers mechanical, electrical and hydraulic parts, not only explaining basic theory and crane configuration, but also combining with port maintenance practice and experience, which is usable and operable. The manual explains the complicated things in a simple way with pictures and illustrations accounting for 50% of the manual contents which is a real illustrated maintenance manual. The manual can act as not only the operation guidance but also the training material for port maintenance personnel.

**If port customers strictly follow the instructions in this manual to do maintenance work correctly and timely, the life time of ZPMC brand high-reliable components can be achieved.**

# About Safety

Ports are the hubs of land and water transportation and also the distributing centers of various cargos. **Port handling operation not only needs the combination of people and machines, but also consists of heavy and complicated tasks which hide many potential hazards.** Therefore, it is the port's obligation to establish relevant safety regulations and procedures to prevent accidents from happening.

The quayside cranes are also potentially hazardous. To reduce the risks of injury or danger, please always follow the instructions in this manual when performing operation and maintenance work on the crane. **Shanghai Zhenhua Heavy Industry Machinery Co. (ZPMC) will not be liable for any damage or injury resulting from abuse or misuse of the crane.**

These quayside cranes are heavy duty, high-speed and very complicated machines with high degree of automation, so it is impossible to foresee every potential hazard. The manufacturer, Shanghai Zhenhua Heavy Industry Machinery Co. (ZPMC), can not envision every possible circumstance that might involve a potential hazard. The warnings in this manual and on the cranes are, therefore, not all inclusive. **When operating, inspecting, or maintaining the cranes, each person must be safety conscious and take all necessary precautions, in addition to following the instructions in this manual.**

# Special Explanation

The <Quayside Container Crane Configuration and Maintenance Manual> is based on the crane that ZPMC has designed and manufactured for CICT in Sri Lanka. It has to be emphasized that some description appeared in this manual are not usual and only possessed by this project. Moreover, ports have accumulated abundant practical experience on their own during many years of crane operation and maintenance. So, **this manual is for reference only.**

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